

ILLINOIS TECH | College of Computing

ITMM 485 / 585
Dr. Gurram Gopal

Legal and Ethical Issues in
Information Technology



1

Ethical Concepts and Theories

ILLINOIS TECH | College of Computing

Ch10: The Digital Divide,
Democracy, And Work
P1: What is the Digital Divide?



2

Announcements

- Internet 02 section- clarification
- Final Exam

ITMM 485/585: Legal and Ethical Issues in IT Slide 1-3

3

Learning Objectives

Upon completion of this lesson the students should be able to:

- Define the **Digital Divide**
 - ❖ Describe and discuss the moral and ethical issues that define and surround the Digital Divide and proposals for its resolution
- Describe and discuss issues surrounding **making technology available to people with disabilities**
- Discuss the effect of **technology on racism**
- Discuss the impact of **technology on gender issues**
 - ❖ Describe and discuss gender issues in employment in the technology field
- Explain and discuss the impact of **technology on democracy and democratic values**
- Describe how technology has **transformed the nature of work and discuss its impact on the quality of work life**

ITMM 485/585: Legal and Ethical Issues in IT Slide 1-4

4

Module Objectives

Upon completion of this lesson the students should be able to:

- Define the **Digital Divide**
 - ❖ Describe and discuss the moral and ethical issues that define and surround the Digital Divide and proposals for its resolution

ITMM 485/585: Legal and Ethical Issues in IT Slide 1-5

5

Access and Equity Issues: The Digital Divide, Democracy, and Work

- 1) **sociodemographic groups** (affecting social/economic class, race, and gender),
- 2) **social and political institutions** (such as education and government),
- 3) **social sectors** (including the workplace).

6

1. Sociodemographic Groups

- The *sociodemographic groups* that have been impacted by cybertechnology include:
- social/economic class (e.g., low-income groups),
- race (i.e., racial and ethnic minority populations),
- gender (women),
- disabled populations.

7

What Is The Digital Divide?

- The phrase “the digital divide,” coined in the 1990s, is a new label for an old concept involving information “haves” and “have-nots” (Compaine, 2001).
- *The digital divide* is now often used to describe the disparity between those who have access to Internet technology and those who do not.

8

Digital Divide

- Benjamin Compaine defines the digital divide as “the gap, or the perceived gap,” between those who have and those who do not have either:
 - a) access to cybertechnology,
 - b) the knowledge and ability to use that technology.



9

Digital Divide

- Kenneth Himma and Maria Bottis (2014) believe that this “divide” or gap is better understood as a “series of gaps” affecting the technological haves and have-nots, which includes gaps (or divisions) between those have and do not have:
 - digital devices and Internet access,
 - the knowledge and ability to use digital tools, and thus enjoy their benefits.



10

Digital Divide

- So, in Himma and Bottis’ view, “the digital divide” does not refer merely to a (single) gap that affects access to information technology; it also affects *knowledge and ability*.
- Along similar lines, Martin Ryder (2015) claims that the digital divide reflects the significant gap between those who can and those who cannot “effectively benefit” from cybertechnology.

11

Digital Divide

- Discussions about *the* digital divide might suggest that there is one overall divide—that is, a single divide as opposed to many “divides,” or divisions.
- There appear to be *multiple divisions* involving access to cybertechnology.
- O’Hara and Stevens (2006) describe three kinds of “divides,” which involve divisions between:
 - rich and poor people,
 - rich and poor regions,
 - rich and poor nations.

12

Digital Divide

- For our purposes, we examine the digital divide as it exists at two distinct levels:
- 1) a "global digital divide" between developed and developing nations.
- 2) a divide between groups within developed nations, based on factors such as income and education.

13

The Global Digital Divide

- In 2000, it was estimated that 429 million people (approximately 6% of the world's population) were online globally.
- 68% of those online in 2000 lived in North America and Europe.
- Two billion people in the world didn't even have electricity, and in developing countries, there were roughly 69 phones for every 1000 people (2000 Human Development Report).

14

Some Statistics from the 2000 Human Development Report

- In 2000, the developing nation of Nepal had
- approximately 35,000 Internet users in a population of 21 million people;
- only 15% of houses with electricity (2000 Human Development Report).

15

Some Statistics Noting Growth Between 2000 and 2005

- Global Internet usage more than doubled between 2000 and 2005.
- In late 2005, it was estimated that there were more than 972 million Internet users (Internet World Stats News, 2005).
- By 2005, seven nations had an Internet penetration rate of higher than 60%, and the list of countries or regions where more than 50% of the population used the Internet had grown to 30 (Internet World Stats News, 2005).

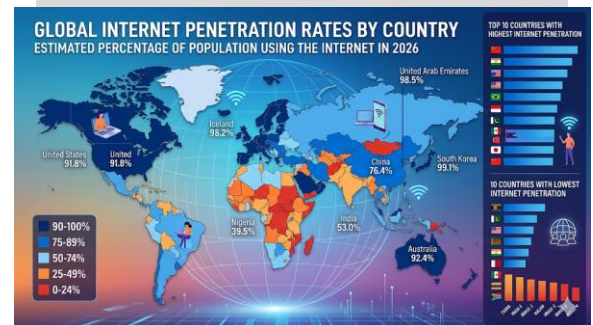
16

Some Statistics Noting Growth Between 2005 and 2011

- In 2011 (World Stats: Usage and Population Studies), it was estimated that there were slightly more than two billion Internet users.
- But despite this increase, the disparity in the percentage of users in developed and developing countries was still significant.
- In India, for example, the penetration rate for Internet users in 2011 was 8.4%, while in the United Kingdom it was 82% that year.

17

Statistical Data



18



Statistical Data

- Initially, one might be encouraged by some reports (from the 2014 statistics) describing growth in Internet usage at the global level, especially in Africa.
- Yet, despite the progress that has been made on the African continent, critics worry that much more work still needs to be done to narrow, and perhaps one day even bridge, the global divide.
- Others worry about the divides that continue to exist *within* nations.

19



The Digital Divide Within Nations

- There are also significant disparities within some developing nations as well.
- In India, for example, the divide appears to be exacerbated for two reasons:
 - 1) a growing segment of the population who are fluent in English and who have the technical literacy required to be able to work in many of the skilled jobs outsourced there are doing very well.
 - 2) those on the other side of the divide – the majority of the population – who have a low level of literacy and little or no access cyberberchnology, are not doing very well.

20



The Digital Divide in the U.S.

- In the U.S., discussions about the digital divide have focused on factors such as:
- income,
- education,
- race,
- Gender
- location.

21



The Digital Divide in the U.S.

- In the 1990s, a National Information Infrastructure (NII) was proposed to ensure that all Americans would have access to information technology.
- The National Telecommunications and Information Administration (NTIA) conducted a series of studies on computer use among various groups.

22



The Digital Divide in the U.S.

- Early NTIA reports confirmed that access to cyberberchnology was related to socio-demographic factors.
- Later NTIA reports have noted that:
 - a) the rate of access to cyberberchnology in the U.S. had increased overall;
 - b) significant disparities still existed with re- spect to socio-demographic factors involving race, education, income, and marital status.

23



Digital Divide in the US: Universal Service vs. Universal Access

- The US Congress passed the Communications Act of 1934, distributing the cost for telephone service to be affordable to all Americans (i.e., providing *universal service*).
- Telephone users still pay a universal connectivity fee, or surcharge, to support universal service.

24

Broadband

For Immediate Release

FCC INCREASES BROADBAND SPEED BENCHMARK

Annual Agency Assessment of High-Speed Internet Service Deployment Establishes New Standard to Better Reflect the Broadband Needs of American Households

WASHINGTON, March 14, 2024—The Federal Communications Commission today adopted its annual assessment of whether advanced telecommunications capability is being deployed in a reasonable and timely fashion across the U.S. In addition to deployment, the Report considers broadband affordability, adoption, availability, and equitable access, when determining whether broadband is being deployed in a reasonable and timely fashion to “all Americans.” The Commission’s Report, issued pursuant to section 706 of the Telecommunications Act of 1996, raises the Commission’s benchmark for high-speed fixed broadband to download speeds of 100 megabits per second and upload speeds of 20 megabits per second – a four-fold increase from the 25/3 Mbps benchmark set by the Commission in 2015.

ITMM 485/585: Legal and Ethical Issues in IT

Slide 1-25

25

Universal Service vs. Universal Access

- Should a universal service policy for the Internet also be subsidized in a similar manner?
- In the case of telephone technology, it was argued that having a telephone was necessary for one’s well being.
- Can the same argument be made for cybertechnology?

26

Universal Service vs. Universal Access

- Universal service policies are controversial because they require subsidies, often resulting either in user fees or higher taxes.
- Advocates of a *universal service* policy for the Internet note that without a government subsidy, people living in less-populated rural areas would not have been able to afford telephone service.
- Critics of universal service argue that Americans already have *universal access* to cybertechnology (e.g., public points of access such as in libraries).

27

Universal Service vs. Universal Access

- Review Scenario 10-2 (in the textbook) on the question of whether in-home Internet service should be provided for public school students in the U.S.
- What is the main rationale or argument used to make the case for providing in-home Internet access to that group?
- Is that argument convincing?

28

The Digital Divide as an *Ethical Issue*

- Is the digital divide an *ethical* issue?
- Is every kind of divide regarding unequal access to goods necessarily an ethical issue?
- For example, skeptics have pointed out that there is a divide between those who have and do not have Mercedes-Benz automobiles, and that many of us fall on the “wrong side” of the “Mercedez-Benz Divide.”

29

The Digital Divide as an Ethical Issue

- Is the divide affecting unequal access to cybertechnology similar to the “Mercedes-Benz Divide”?
- Or, is it closer to divisions involving access to vital human resources such as food and healthcare?

30



The Digital Divide as an Ethical Issue

- Many ethicists believe that divisions between those who do and do not have access to vital resources, such as food and healthcare, raise questions of *distributive justice*.
- Distributive justice refers to the “just distribution” of primary goods and resources in populations.

31



Distributive Justice

- What, exactly, do we mean by “distributive justice” in the context of cybertechnology?
- According to van den Hoven and Rooksby (2008):
Distributive justice in contemporary information societies concerns, among other issues, the distribution of information, information services, and information infrastructures. [Italics Added]
- Van den Hoven and Rooksby’s view is influenced by Rawls’ theory of justice and “primary social goods.”

32



Distributive Justice and Rawls

- In his classic work, *A Theory of Justice*, John Rawls describes *primary social goods* as “resources that satisfy basic human needs and thus have a special value or moral weight in society.”
- Rawls notes that with these goods, humans “can generally be assured of greater success in carrying out their intentions and in advancing their needs.”
- Van den Hoven and Rooksby extend Rawls’ notion of primary social goods to include “information goods.”

33



Bridging the Digital Divide

- Moss (2002) argues that people in developing countries who do not have access to cybertechnology are unfairly disadvantaged because:
 - i. they are denied access to knowledge;
 - ii. they are unable to participate fully in democratic decision making processes;
 - iii. their prospects for economic growth are hindered.

34



Bridging the Digital Divide

- Norris (2001) believes that “the underclass of the “information poor” may become “further marginalized in societies where basic computer skills are becoming essential for economic success and personal advancement.”
- Norris also notes that these skills are necessary for “entry to good career and educational opportunities,” as well as “full access to social networks...and opportunities for civic engagement.”

35



Do We Have a Moral Obligation to Bridge the Digital Divide?

- If Moss and Norris are correct, it would seem that we have a moral obligation to provide access to those who are disadvantaged?
- However, some argue that we are morally obligated *only* to “do no harm”
- For example, they believe we have no explicit “obligation to do good” — in this case, no obligation to provide Internet access to disadvantaged groups.
- But we can question whether a “minimalist” view of morality is adequate (see Chapter 2).

36

Do We Have a Moral Obligation to Bridge the Digital Divide?

- Bottis and Himma (2008) note that some important points of clarification are needed to make the case that “affluent nations” have a moral obligation to bridge the divide.
- They also point out that we need to draw an important distinction between saying that:
- “X is a good thing to do.”
- “We are obligated to do X.”

37

Do We Have a Moral Obligation to Bridge the Divide ?

- Bottis and Himma believe that most people would likely agree that eliminating the digital divide would be “a good thing to do.”
- But they also note that there would likely be far less consensus as to whether we – i.e., some affluent nations – have an obligation to do it.

38

Chicago Public Libraries

Spotlight on >



LinkedIn Learning

Adults Technology



Morningstar

Adults personal finance



Cell-Ed

Adults English as a Second Language language learning literacy learning



Mango Languages

Adults Kids Teens Teachers language learning

ITMM 485/585: Legal and Ethical Issues in IT

Slide 1-39

39

Do We Have a Moral Obligation to Bridge the Divide ?

- Himma (2007) points out that because many people believe that we are morally obligated *only* to do no harm, those people tend to infer that we have no moral obligation to bridge the digital divide.
- But Himma also argues that this view is “inconsistent with the ethics of every classically theistic religion as well as our ordinary intuitions,” as well as with most ethical theories.

40